Sense & Stop

Member 4 (forward sensing and EEG data): what Member 5 needs to know

30 October 2026 | Interface note: `docs/interface_note_forward.md` in `Jana-Hazem/sense-and-stop` (merged)

In one paragraph. Member 4 built the forward map that turns Member 3's hidden wave u into simulated scalp EEG: **y = H A u + ε**. A flat 30 × 20 mm cortical patch on the fsaverage template receives the wave through an area-averaging matrix A; a lead-field operator H turns the source values into 21 bipolar EEG channels; ε is real background EEG taken from seizure-free CHB-MIT recordings and reduced to the PDE time grid. This document lists what was built, what files you receive, and the limits you must respect when you invert it.

Scope. This is a computational study. The wave comes from a simplified model tuned to rat data; the noise is pediatric epilepsy EEG; the head is an adult template. None of these is a CSD or migraine recording of the same subject, and nothing here is a clinical result.

1. The forward model in brief

Item	Value
Equation	y = H A u + ε. In code the sensor array is called `sensors` (not y, because y is a spatial axis in the PDE solver).
Input u	Member 3's default 2D run, `u[t, y, x]`, shape 271 × 201 × 301 (about 131 MB in the default run; the copy you receive is about 65 MB). Units: mm and s. dx = 0.1 mm, output every 2 s, T = 540 s, patch 30 mm (x) by 20 mm (y).
Source space	Flat patch on `lateraloccipital-lh` of the MNE fsaverage template. Placement on occipital cortex is a design choice, not data-based.
A (PDE → sources)	Sparse area-averaging matrix; the final patch has **73 sources**. (An earlier flattening step gave a count of 137; the note records 73 as the final 30 × 20 mm cut.)
H_bip (sources → channels)	Bipolar lead field, units **V per A·m**. Orientation fixed to the cortical normal. Each channel is **first electrode minus second**.
Output	`sensors[271, 21]`: 271 time points (2 s apart) by 21 bipolar channels, in volts.
Scale u → volts	**Illustrative.** u is dimensionless, so one scale factor converts it to current density. Three scales are provided: 1e-9, 1e-8, 1e-7. Report results at all three.

2. The 21 channels (fixed order)

`FP1-F7, F7-T7, T7-P7, P7-O1, FP1-F3, F3-C3, C3-P3, P3-O1, FP2-F4, F4-C4, C4-P4, P4-O2, FP2-F8, F8-T8, T8-P8, P8-O2, FZ-CZ, CZ-PZ, T7-FT9, FT9-FT10, FT10-T8.`

The duplicate T8-P8 channel and the sign-flipped P7-T7 channel in the CHB-MIT files were removed. The channel order of `H_bip.npz`, `noise_2s.npz` and `noise_cov.npz` was checked to match.

3. Files you receive (sent outside Git)

File	What it is
H_bip.npz	Bipolar lead field and channel names.
A_matrix.npz	Sparse area-averaging matrix (PDE nodes → 73 sources).
noise_2s.npz	Real background EEG, volts, shape [3 files, 21 channels, 1740 samples], dt = 2 s.
noise_cov.npz	Noise covariance C (21 × 21) with channel names.
sensors_default.npz	Default synthetic observation, including `clean_V` [271, 21] in volts.
sensors.py	Contains `make_sensors`, which builds sensor data from u at a chosen scale.
u_2d_default.npy	Member 3's default 2D wave (about 65 MB).

No .npy or .npz file is in the repository. The scripts `build_A.py`, `build_Hbip.py`, `make_noise.py` and `sensors.py` are in `src/forward/` but are provisional: they contain local `D:\` paths and will be made path-configurable later. Do not rerun `build_Hbip.py`, because it overwrites H_bip.npz.

4. How the noise was made

- Source: CHB-MIT case chb01, files chb01_01, chb01_02, chb01_08. All three have 0 seizures per the summary file.
- Anti-aliased decimation from 256 Hz to 0.5 Hz (2 s grid) in three stages of 8, zero-phase, then 60 s trimmed from each end to drop filter edge effects. Each file gives 1740 samples (3480 s).
- Decimation removes most normal EEG content (the decimated standard deviation is about 9% to 26% of the raw one). The noise is therefore **empirical, correlated across channels and band-limited**, not white.
- Decimated standard deviation per channel: median 3.77 µV, maximum 10.48 µV (per-file mean). The left occipital-parietal channels are about 2.1 to 2.4 µV. The right channels P4-O2 and P8-O2 are much noisier: 8.7 and 10.5 µV by per-file mean, 10.8 and 12.0 µV in the pooled covariance `noise_cov.npz`.
- **Slow drift** (18 windows of 540 s): the linear trend over 540 s has median 0.02 to 0.06 µV and 95th percentile 0.09 to 0.20 µV on the left channels, and 1.3 to 1.5 µV on P8-O2 and P4-O2. A linear trend explains under 0.5% of the variance. This is small at scales 1e-7 and 1e-8 but not at 1e-9, where the peak clean signal is only 0.13 to 0.58 µV.
- The EDF prefilter field (read from the chb01_01 header only) is empty. That does not mean the data are unfiltered, so it is unknown whether very slow drift was removed before storage. If it was, real drift is under-represented here.

5. Rules for the inversion

- **Do not clip u.** The excitable model undershoots to about −0.285 behind the pulse. Clipping to [0, 1] is wrong.
- **Do not reuse the same discretisation** for inversion and for data generation (avoids the inverse crime). Member 3's RK4 and scipy solve_ivp solvers agree to below 5e-3, so use a different grid or solver.
- **Cut or flag the first ~60 s.** That window is initiation of the wave from the stimulus.
- **Stay away from x = 30 mm.** Values near the far edge are unreliable because P7-O1 flips sign there. The default run's front reaches only about 25.6 mm by 540 s.
- **The default stimulus is a full-width strip.** At t = 0, u = 1.0 at x = 0 for every y from 0 to 20 mm, and the region above 0.5 covers x = 0 to 2.9 mm and y = 0 to 20 mm. So u is exactly uniform in y and this run **cannot test y-structure**. The wave is effectively planar.
- **Report results at all three scales** (1e-9, 1e-8, 1e-7). The scale is illustrative, so a conclusion that holds at only one scale should not be presented as general.

6. What the operator can and cannot resolve

- **Rank.** The matrix has at most 19 independent directions by construction (21 channels minus 2 loops). The 19th is at float32 noise level, so treat it as effectively 18 or fewer. Usable directions after noise are fewer still; the interface note gives the current wording on noise-aware rank.
- **Localisation.** The channel pattern barely changes when the front moves 1 to 5 mm (similarity 0.975 or higher). Position therefore has to come from **timing and amplitude**, not pattern shape. This is a property of the operator before any noise is added.
- **FT9 and FT10.** These rows come from the same forward pipeline but were not independently checked. The loop check cancels them algebraically, so it cannot detect an error in them.
- **SNR.** Quote only late-window values (t ≥ 100 s) and only as upper bounds. The right-side channels have SNR of about 1.9. The note's section 7 table gives the exact numbers per scale (for example P3-O1 peaks at 42.6 µV at scale 1e-7); where an earlier informal list differs, the note takes precedence.

7. What Member 3's wave looks like (limits inherited from the model)

Feature	Data (rat, 3 animals)	Default model
Pulse duration (FWHM)	42.8 s	42.8 s (calibration target)
Front speed	2.3 to 2.9 mm/min (literature, not verified)	2.60 mm/min
Pulse width	1.7 to 2.1 mm (derived)	1.85 mm
Decay time (peak to 10%)	40.7 s	**28.2 s**: the model recovers too fast, so the pulse tail at the sensors is a bit short
Wide-pulse scenario	targets 6.4 mm at 9 mm/min	simulated **6.71 mm at 9.27 mm/min** (1D only; the 2D versions were not run)

Model: excitable FitzHugh-Nagumo type, D = 0.0186 mm^2/s, k = 0.504 s^{-1}. The grid spacing is 0.1 mm, with under 2% change in the convergence test.

8. Assumptions and open items

- The head is an adult template (fsaverage) and the EEG is pediatric (ages 1.5 to 22). Source locations must not be over-interpreted (decision D4).
- The license of the S3 template data files is unconfirmed. The CHB-MIT file checksums were not verified against SHA256SUMS.txt.
- The u-to-volts scale is illustrative. Absolute amplitudes in volts are not physiological predictions.
- Rat data have one electrode only, so wave speed comes from literature and the 2D geometry is a design choice.

9. Suggested first steps

- Read sections 7 to 9 of `docs/interface_note_forward.md` (SNR, rank, drift, limits).
- Load `sensors_default.npz` and check that `sensors` is [271, 21] with the channel order above.
- Generate your own noisy data with a different discretisation, run the inversion at all three scales, and compare.